CAR rCARI Validation Study Primary Results

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1 Survey Progress

Table 1: Sample Completion: Face-to-Face vs. Remote

	Not Completed		Completed		Total	
	Count	Percent	Count	Percent	Count	Percent
Face-To-Face	295	23.34	969	76.66	1264	100.00
Remote	746	72.92	277	27.08	1023	100.00
Total	1041	45.52	1246	54.48	2287	100.00

Figure 1: Data Collection Progress by Date and Modality

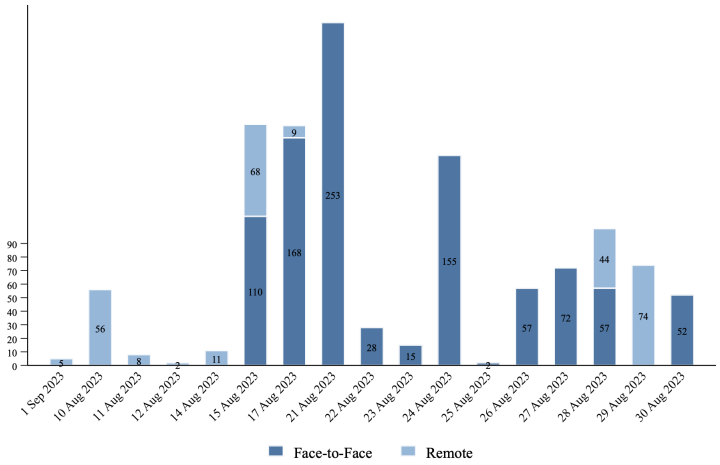


Figure 2: F2F Data Collection Progress by Date and Enumerator

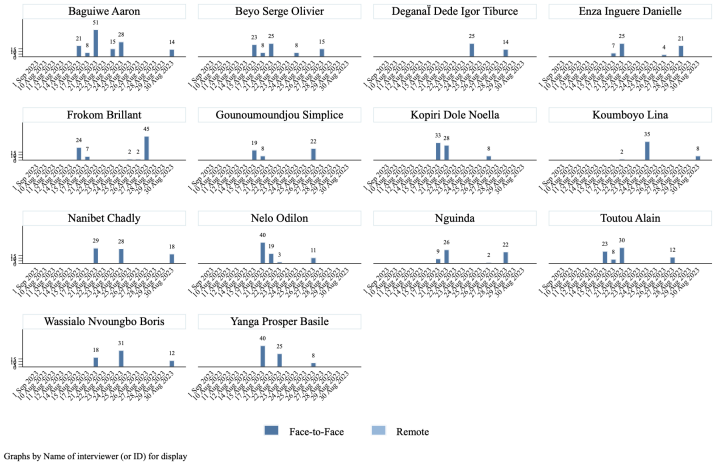
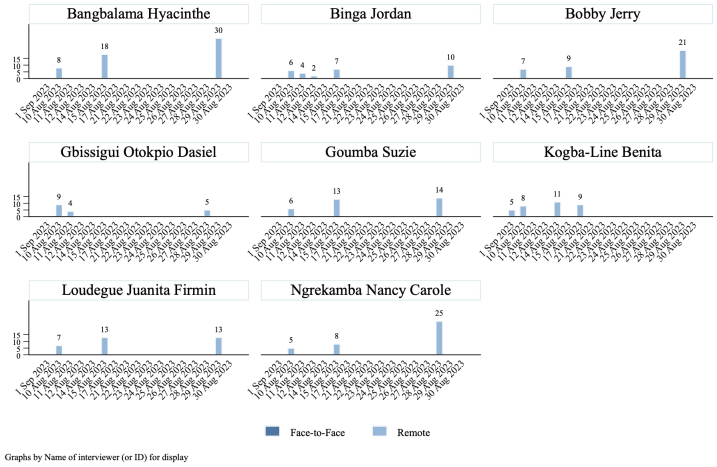


Figure 3: Remote Data Collection Progress by Date and Enumerator



2 Randomization Quality

To assess the balance of the treatment (Remote) and control (F2F) groups, we examined various respondent and household characteristics, including basic demographics, income sources, and changes over time. This allowed us to evaluate whether the two groups were similar in terms of these factors, ensuring a fair comparison between the treatment and control conditions.

2.1 Modality Randomization

Table 2: Balance Table on Randomized Survey Modality Assignment: Face-to-Face vs. Remote

		(1) F2F		(2) Remote		(1)-(2) Pairwise t-test
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Respondent Age	969	37.80	277	37.67	1246	0.13
		(0.44)		(0.74)		
Female Respondent	969	0.54	277	0.43	1246	0.11***
		(0.02)		(0.03)		
Head of Household Respondent	969	0.64	277	0.70	1246	-0.06*
		(0.02)		(0.03)		
Household Size	969	6.14	277	7.71	1246	-1.57***
		(0.08)		(0.19)		
How many household members are contributing to the household income?	955	1.76	277	1.66	1232	0.10
		(0.03)		(0.05)		
How many income sources does your household have?	969	1.50	277	1.47	1246	0.03
		(0.02)		(0.04)		
Income Source: None	969	0.01	277	0.00	1246	0.01
		(0.00)		(0.00)		
Income Source: Skilled Waged Labor	969	0.05	277	0.09	1246	-0.04***
		(0.01)		(0.02)		
Income Source: Unskilled Waged Labor	969	0.14	277	0.35	1246	-0.21***
		(0.01)		(0.03)		
Income Source: Trade	969	0.40	277	0.18	1246	0.22***
		(0.02)		(0.02)		
Income Source: Agricultural Production	969	0.27	277	0.34	1246	-0.07**
		(0.01)		(0.03)		
Income Change: None	969	0.54	277	0.35	1246	0.19***
		(0.02)		(0.03)		
Income Change: Increase	969	0.06	277	0.16	1246	-0.10***
		(0.01)		(0.02)		
Income Change: Reduced < 25%	969	0.14	277	0.28	1246	-0.14***
		(0.01)		(0.03)		
Income Change: Reduced 25% - 50%	969	0.17	277	0.13	1246	0.04
		(0.01)		(0.02)		
Income Change: Reduced 50+%	969	0.09	277	0.08	1246	0.02
		(0.01)		(0.02)		

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

2.2 F2F Length Randomization

Table 3: Balance Table on Randomized Survey Length Assignment in F2F Survey: Long vs. Short

Variable	N	(1) F2F Mean/(SE)	N	(2) Remote Mean/(SE)	N	(1)-(2) Pairwise t-test Mean difference
Respondent Age	466	37.30	503	38.27	969	-0.97
		(0.60)		(0.64)		
Female Respondent	466	0.54	503	0.53	969	0.01
		(0.02)		(0.02)		
Head of Household Respondent	466	0.64	503	0.63	969	0.01
		(0.02)		(0.02)		
Single Respondent	466	0.22	503	0.22	969	-0.00
		(0.02)		(0.02)		
Married Respondent	466	0.06	503	0.09	969	-0.03*
		(0.01)		(0.01)		
IDP Household	466	0.00	503	0.00	969	-0.00
		(0.00)		(0.00)		
Household has disabled member	466	0.24	503	0.22	969	0.02
		(0.02)		(0.02)		
Number of disabled household members	466	0.29	503	0.27	969	0.02
		(0.03)		(0.03)		
Household head has attended school	374	0.70	423	0.67	797	0.03
		(0.02)		(0.02)		
Household Size	466	6.12	503	6.16	969	-0.04
		(0.12)		(0.11)		
Total number of male household member	466	2.83	503	2.83	969	-0.00
		(0.08)		(0.07)		
Total number of female household member	466	2.95	503	2.99	969	-0.05
		(0.08)		(0.07)		
Total number of children	466	3.71	503	3.71	969	-0.01
		(0.10)		(0.09)		
Total number of senior	466	0.50	503	0.50	969	0.00
		(0.05)		(0.04)		
Total number of adults	466	1.94	503	1.95	969	-0.01
		(0.06)		(0.05)		
Household Female/Male Ratio	466	1.54	503	1.61	969	-0.07
		(0.07)		(0.07)		
Household Dependency Ratio	466	3.01	503	2.97	969	0.03
		(0.12)		(0.12)		
How many household members are contributing to the household income?	462	1.74	493	1.78	955	-0.04
		(0.04)		(0.05)		
How many income sources does your household have?	466	1.51	503	1.49	969	0.02
		(0.03)		(0.03)		
Income Source: None	466	0.01	503	0.01	969	0.00
		(0.00)		(0.00)		
Income Source: Skilled Waged Labor	466	0.05	503	0.04	969	0.01
		(0.01)		(0.01)		
Income Source: Unskilled Waged Labor	466	0.12	503	0.16	969	-0.04*
		(0.01)		(0.02)		
Income Source: Trade	466	0.43	503	0.38	969	0.05
		(0.02)		(0.02)		
Income Source: Agricultural Production	466	0.26	503	0.28	969	-0.02
		(0.02)		(0.02)		
Income Change: None	466	0.55	503	0.53	969	0.02
		(0.02)		(0.02)		
Income Change: Increase	466	0.06	503	0.06	969	0.00
		(0.01)		(0.01)		
Income Change: Reduced < 25%	466	0.14	503	0.15	969	-0.01
		(0.02)		(0.02)		
Income Change: Reduced 25% - 50%	466	0.17	503	0.17	969	0.00
		(0.02)		(0.02)		
Income Change: Reduced 50+%	466	0.09	503	0.10	969	-0.01
		(0.01)		(0.01)		

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

2.3 Remote Length Randomization

Table 4: Balance Table on Randomized Survey Length Assignment in Remote Survey: Long vs. Short

Variable	N	(1) F2F Mean/(SE)	N	(2) Remote Mean/(SE)	N	(1)-(2) Pairwise t-test Mean difference
Respondent Age	120	38.01	157	37.41	277	0.59
		(1.16)		(0.97)		
Female Respondent	120	0.39	157	0.46	277	-0.07
		(0.04)		(0.04)		
Head of Household Respondent	120	0.71	157	0.69	277	0.02
		(0.04)		(0.04)		
Household Size	120	7.49	157	7.88	277	-0.39
		(0.27)		(0.26)		
How many household members are contributing to the household income?	120	1.59	157	1.71	277	-0.12
		(0.07)		(0.06)		
How many income sources does your household have?	120	1.41	157	1.51	277	-0.10
		(0.05)		(0.06)		
Income Source: None	120	0.00	157	0.00	.n	.n
		(0.00)		(0.00)		
Income Source: Skilled Waged Labor	120	0.07	157	0.11	277	-0.04
		(0.02)		(0.02)		
Income Source: Unskilled Waged Labor	120	0.32	157	0.38	277	-0.06
		(0.04)		(0.04)		
Income Source: Trade	120	0.18	157	0.18	277	-0.00
		(0.04)		(0.03)		
Income Source: Agricultural Production	120	0.40	157	0.29	277	0.11*
		(0.04)		(0.04)		
Income Change: None	120	0.33	157	0.37	277	-0.04
		(0.04)		(0.04)		
Income Change: Increase	120	0.15	157	0.17	277	-0.02
		(0.03)		(0.03)		
Income Change: Reduced < 25%	120	0.33	157	0.25	277	0.08
		(0.04)		(0.03)		
Income Change: Reduced 25% - 50%	120	0.10	157	0.16	277	-0.06
		(0.03)		(0.03)		
Income Change: Reduced 50+%	120	0.10	157	0.06	277	0.04
		(0.03)		(0.02)		

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

3 Primary Results

3.1 Impact of Survey Modality on Key Food Security Outcomes

Table 5: Impact of Survey Modality: Face-to-Face vs. Remote

Variable	(1) F2F Mean/(SE)	(2) Remote Mean/(SE)	(1)-(2) Pairwise t-test Mean difference
Consumption over the past 7 days (cereals and tubers)	5.32	5.65	-0.33**
	(0.07)	(0.11)	
Consumption over the past 7 days (pulses)	2.54	2.99	-0.45***
	(0.07)	(0.13)	
Consumption over the past 7 days (dairy products)	0.20	0.82	-0.62***
	(0.03)	(0.11)	
Consumption over the past 7 days (protein-rich foods)	0.88	1.45	-0.58***
	(0.05)	(0.10)	
Consumption over the past 7 days (vegetables)	3.99	5.61	-1.62***
	(0.09)	(0.10)	
Consumption over the past 7 days (fruit)	0.38	0.67	-0.29***
	(0.04)	(0.08)	
Consumption over the past 7 days (oil)	2.87	4.49	-1.62***
	(0.09)	(0.15)	
Consumption over the past 7 days (sugar)	1.62	2.82	-1.20***
	(0.07)	(0.16)	
Food Consumption Score	29.19	39.31	-10.12***
	(0.51)	(1.09)	
Relied on less preferred, less expensive food	3.55	2.98	0.57***
	(0.07)	(0.13)	
Borrowed food or relied on help from friends or relatives	0.81	0.82	-0.01
	(0.04)	(0.07)	
Reduced portion size of meals at meals time	2.00	2.57	-0.57***
	(0.08)	(0.13)	
Restricted consumption by adults in order for young-children to eat	1.58	2.58	-0.99***
	(0.06)	(0.12)	
Reduced the number of meals eaten per day	1.61	2.12	-0.51***
	(0.08)	(0.13)	
Reduced Consumption Strategies Index (rCSI)	13.52	17.04	-3.51***
	(0.34)	(0.56)	
Household engaged in stress coping strategies	0.62	0.81	-0.19***
	(0.02)	(0.02)	
Household engaged in crisis coping strategies	0.15	0.35	-0.20***
	(0.01)	(0.03)	
Household engaged in emergency coping strategies	0.05	0.23	-0.17***
	(0.01)	(0.03)	
Total monthly food expenditure (average)	28,510.90	50,565.32	-22,054.41***
	(791.17)	(3,276.90)	
Total monthly non-food expenditure (average)	11,816.76	30,637.90	-18,821.14***
	(503.80)	(9,067.01)	
Monthly total household expenditure	40,327.66	81,203.22	-40,875.56***
	(1,079.77)	(9,978.36)	
Monthly total household expenditure per capita	7,327.98	11,163.12	-3,835.15***
	(258.59)	(1,215.31)	
Food Expenditure Share	0.71	0.67	0.04***
	(0.01)	(0.02)	
Household being vulnerable using FES CARI	0.86	0.76	0.10***
	(0.01)	(0.03)	
Household being vulnerable using Income rCARI	0.67	0.63	0.04
	(0.02)	(0.03)	
Number of observations	969	277	1246

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table 6: Impact of Survey Length in F2F Survey: Long vs. Short

Variable	(1) Total Mean/(SE)	(2) Long Mean/(SE)	(3) Short Mean/(SE)	(2)-(3) Pairwise t-test Mean difference
Consumption over the past 7 days (cereals and tubers)	5.32	5.30	5.34	-0.04
	(0.07)	(0.10)	(0.09)	
Consumption over the past 7 days (pulses)	2.54	2.58	2.50	0.08
	(0.07)	(0.10)	(0.10)	
Consumption over the past 7 days (dairy products)	0.20	0.23	0.17	0.05
	(0.03)	(0.05)	(0.04)	
Consumption over the past 7 days (protein-rich foods)	0.88	0.84	0.91	-0.07
	(0.05)	(0.06)	(0.07)	
Consumption over the past 7 days (vegetables)	3.99	2.79	5.11	-2.32***
	(0.09)	(0.14)	(0.09)	
Consumption over the past 7 days (fruit)	0.38	0.35	0.41	-0.06
	(0.04)	(0.05)	(0.05)	
Consumption over the past 7 days (oil)	2.87	2.80	2.93	-0.13
	(0.09)	(0.12)	(0.12)	
Consumption over the past 7 days (sugar)	1.62	1.61	1.63	-0.02
	(0.07)	(0.10)	(0.10)	
Food Consumption Score	29.19	27.98	30.31	-2.34**
	(0.51)	(0.72)	(0.73)	
Relied on less preferred, less expensive food	3.55	3.54	3.56	-0.02
	(0.07)	(0.11)	(0.10)	
Borrowed food or relied on help from friends or relatives	0.81	0.84	0.78	0.06
	(0.04)	(0.06)	(0.06)	
Reduced portion size of meals at meals time	2.00	2.07	1.93	0.14
	(0.08)	(0.11)	(0.10)	
Restricted consumption by adults in order for young-children to eat	1.58	1.62	1.55	0.07
	(0.06)	(0.09)	(0.08)	
Reduced the number of meals eaten per day	1.61	1.70	1.52	0.18
	(0.08)	(0.11)	(0.11)	
Reduced Consumption Strategies Index (rCSI)	13.52	13.86	13.21	0.64
	(0.34)	(0.49)	(0.46)	
Household engaged in stress coping strategies	0.62	0.63	0.62	0.00
	(0.02)	(0.02)	(0.02)	
Household engaged in crisis coping strategies	0.15	0.15	0.15	0.01
	(0.01)	(0.02)	(0.02)	
Household engaged in emergency coping strategies	0.05	0.05	0.05	0.00
	(0.01)	(0.01)	(0.01)	
Total monthly food expenditure (average)	28,510.90	31,018.90	26,187.39	4,831.50***
	(791.17)	(1,286.05)	(939.97)	
Total monthly non-food expenditure (average)	11,816.76	12,763.36	10,939.79	1,823.57*
	(503.80)	(779.15)	(647.07)	
Monthly total household expenditure	40,327.66	43,782.26	37,127.18	6,655.07***
	(1,079.77)	(1,742.37)	(1,297.42)	
Monthly total household expenditure per capita	7,327.98	8,054.64	6,654.77	1,399.87***
	(258.59)	(467.89)	(242.17)	
Food Expenditure Share	0.71	0.71	0.72	-0.01
	(0.01)	(0.01)	(0.01)	
Household being vulnerable using FES CARI	0.86	0.87	0.84	0.03
	(0.01)	(0.02)	(0.02)	
Household being vulnerable using Income rCARI	0.67	0.69	0.66	0.03
	(0.02)	(0.02)	(0.02)	
Number of observations	969	466	503	969

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table 7: Impact of Survey Length in Remote Survey: Long vs. Short

Variable	(1) Total Mean/(SE)	(2) Long Mean/(SE)	(3) Short Mean/(SE)	(2)-(3) Pairwise t-test Mean difference
Consumption over the past 7 days (cereals and tubers)	5.65	5.83	5.51	0.32
	(0.11)	(0.17)	(0.15)	
Consumption over the past 7 days (pulses)	2.99	3.07	2.93	0.14
	(0.13)	(0.20)	(0.16)	
Consumption over the past 7 days (dairy products)	0.82	0.69	0.92	-0.23
	(0.11)	(0.15)	(0.16)	
Consumption over the past 7 days (protein-rich foods)	1.45	1.11	1.72	-0.61***
	(0.10)	(0.14)	(0.14)	
Consumption over the past 7 days (vegetables)	5.61	5.49	5.71	-0.22
	(0.10)	(0.16)	(0.13)	
Consumption over the past 7 days (fruit)	0.67	0.74	0.61	0.13
	(0.08)	(0.13)	(0.09)	
Consumption over the past 7 days (oil)	4.49	4.08	4.81	-0.73**
	(0.15)	(0.22)	(0.20)	
Consumption over the past 7 days (sugar)	2.82	2.51	3.06	-0.56*
	(0.16)	(0.24)	(0.21)	
Food Consumption Score	39.31	37.60	40.61	-3.02
	(1.09)	(1.43)	(1.58)	
Relied on less preferred, less expensive food	2.98	3.14	2.86	0.28
	(0.13)	(0.19)	(0.17)	
Borrowed food or relied on help from friends or relatives	0.82	0.76	0.87	-0.11
	(0.07)	(0.10)	(0.10)	
Reduced portion size of meals at meals time	2.57	2.58	2.55	0.03
	(0.13)	(0.20)	(0.16)	
Restricted consumption by adults in order for young-children to eat	2.58	2.89	2.34	0.55**
	(0.12)	(0.19)	(0.15)	
Reduced the number of meals eaten per day	2.12	2.50	1.82	0.68***
	(0.13)	(0.20)	(0.16)	
Reduced Consumption Strategies Index (rCSI)	17.04	18.42	15.98	2.44**
	(0.56)	(0.87)	(0.71)	
Household engaged in stress coping strategies	0.81	0.75	0.86	-0.11**
	(0.02)	(0.04)	(0.03)	
Household engaged in crisis coping strategies	0.35	0.33	0.37	-0.04
	(0.03)	(0.04)	(0.04)	
Household engaged in emergency coping strategies	0.23	0.23	0.23	-0.00
	(0.03)	(0.04)	(0.03)	
Total monthly food expenditure (average)	50,565.32	40,460.82	58,288.50	-17,827.68***
	(3,276.90)	(4,071.12)	(4,793.00)	
Total monthly non-food expenditure (average)	30,637.90	19,262.26	39,332.66	-20,070.40
	(9,067.01)	(2,583.85)	(15,862.22)	
Monthly total household expenditure	81,203.22	59,723.08	97,621.16	-37,898.08*
	(9,978.36)	(5,870.92)	(16,932.70)	
Monthly total household expenditure per capita	11,163.12	8,827.35	12,948.42	-4,121.07*
	(1,215.31)	(856.92)	(2,033.49)	
Food Expenditure Share	0.67	0.56	0.76	-0.19***
	(0.02)	(0.03)	(0.02)	
Household being vulnerable using FES CARI	0.76	0.71	0.80	-0.09*
	(0.03)	(0.04)	(0.03)	
Household being vulnerable using Income rCARI	0.63	0.62	0.63	-0.01
	(0.03)	(0.04)	(0.04)	
Number of observations	277	120	157	277

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

4 Outlier Checks

Table 8: Outlier Check for FCS

	Face-to-Face				Remote			
	Mean	SD	Range	Coef of Variation	Mean	SD	Range	Coef of Variation
Food Consumption Score:								
FCSStap	5.32	2.11	7	0.40	5.65	1.85	6	0.33
FCSPulse	2.54	2.20	7	0.87	2.99	2.08	7	0.70
FCSDairy	0.20	0.99	7	4.94	0.82	1.89	7	2.31
FCSPr	0.88	1.45	7	1.64	1.45	1.69	7	1.16
FCSVeg	3.99	2.81	7	0.70	5.61	1.71	7	0.30
FCSFruit	0.38	1.13	7	2.95	0.67	1.29	7	1.93
FCSFat	2.87	2.73	7	0.95	4.49	2.49	7	0.55
FCSSugar	1.62	2.26	7	1.39	2.82	2.60	7	0.92
FCS	29.19	15.94	98	0.55	39.31	18.14	88	0.46
Reduced Coping Strategy Index:								
rCSILessQlty	3.55	2.32	7	0.65	2.98	2.13	7	0.72
rCSIBorrow	0.81	1.23	7	1.52	0.82	1.16	6	1.41
rCSIMealSize	2.00	2.35	7	1.18	2.57	2.12	7	0.83
rCSIMealAdult	1.58	1.83	7	1.16	2.58	1.98	7	0.77
rCSIMealNb	1.61	2.41	7	1.50	2.12	2.09	7	0.99
rCSI	13.52	10.49	56	0.78	17.04	9.27	46	0.54
Livelihood Coping Strategies:								
Lcs_Stress_Coping	0.62	0.48	1	0.78	0.81	0.39	1	0.48
Lcs_Crisis_Coping	0.15	0.36	1	2.37	0.35	0.48	1	1.35
Lcs_Emergency_Coping	0.05	0.22	1	4.24	0.23	0.42	1	1.85
Expenditure:								
HHExpFoodTotal_1M	28510.90	24628.11	199247	0.86	50565.32	54538.43	443571	1.08
HHExpNFTotal_1M	11816.76	15682.67	166167	1.33	30637.90	150905.07	2003008	4.93
HHExpTotal	40327.66	33611.78	255819	0.83	81203.22	166072.96	2101580	2.05
PCExpTotal	7327.98	8049.47	167683	1.10	11163.12	20226.79	224010	1.81
FES	0.71	0.18	1	0.25	0.67	0.28	1	0.42
HH_Vul_FES	0.86	0.35	1	0.41	0.76	0.43	1	0.57
Income:								
HH_Vul_Income	0.67	0.47	1	0.70	0.63	0.48	1	0.77